

04 — DP Redundancy 400 kW (the RESERVE function)

Two modes: Transit 5 000 h (11 463/3 800, NO battery there) and DP 2 000 h (thrust 2 500, hotel 1 500, redundancy 400). Battery 500 kW, **DP only**.

Battery Contribution (DP cascade, budget 500)

Row	H	I	J	L	Note
DpReserve (RESERVE, D=1.0)	400 (the FULL requirement)	400	400	0	covered kW-for-kW, zero factor loss
DpDemand (D=0.5, vf=0)	2 500×0 = 0	0	0	0	thrusters have no swing in the model
Hotel (D=0.05)	1 500×2 % = 30	30	1.5	28.5	budget left: 70

Tiles: **SR = 28.5 · PS = 1.5** (PS counts peak-shaving rows only; the reserve's 400 is full coverage, not a ± band — Excel's J10 would show 401.5).

Power Demands (two rows per table now)

- Transit: propulsion **11 463** (clean!), hotel 3 800 = SG 3 250 + AE **550**
- DP: propulsion **2 500**, hotel 1 500+28.5 = **1 529** — all via SG (the 400 is NOT in the demand: a reserve is readiness, not load)
- Energy: Transit ME (11 463+3 250)×5 000 = **73 565 000** · DP ME (2 500+1 528.5)×2 000 = **8 057 000** · AE 550×5 000 = **2 750 000**
- Header = hours-weighted: ME (73 565 000+8 057 000)/7 000 = **11 660** · AE 2 750 000/7 000 = **393**

Baseline — two modes, two different rules

- Transit (no battery there) → default = last row: 2 ME+SG, 2.6255 → 13 127.5
- DP (battery active) → 3rd-from-worst clamped: only 2 combos ⇒ index 0 = its own optimal, 0.70385×2 000 = 1 407.7
- Baseline total = **14 535 t/yr**

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Transit optimal 2.5785×5 000 = 12 892.5 + DP 1 407.7 = **14 300.2 t/yr**. Savings = **234.6 t = \$182 996** — ALL from Transit (DP baseline = DP optimal after the clamp).

Battery Benefit

DP only: the twin must spin 430 kW of reserve (400+30) vs our 28.5. Benefit = $\Delta\text{FOC} \times 2\,000\text{ h} = \mathbf{139.9\text{ t/yr} = \$109\,113}$.

Takeaway: RESERVE $\neq$ PEAK SHAVING. A reserve wants the full amount, is covered 100 %, adds nothing to demand when covered — the battery's strongest use-case (a class requirement satisfied with zero extra spinning iron).